

# Systems of Linear Equations :

## Systems of Linear Equations

### \* Table of Content

- ① Introduction to Linear Equation in 1, 2, 3 -- n variable
- ② Introduction to system of Linear equations
- ③ Row picture and column picture for systems of Linear equations
- ④ Machine Learning usecase for systems of Linear equations

### \* Linear equation

→ An equation where the highest power of variable is 1.

① Linear equation in 1 variable

$$ax + b = 0 \ (a \neq 0) \leftarrow \text{Represents a point}$$

Example: ①  $5x + 30 = 0$

②  $2x - 3 = 0$

→  $5x + 30 = 0$

$$x = -30/5 = -6$$

$2x - 3 = 0$

$$x = 3/2 = 1.5$$



② Linear equation in 2 variables

$$ax + by + c = 0 \leftarrow \text{Represents a straight line in } 2D(\mathbb{R}^2)$$

Example: ①  $x + 2y - 3 = 0$

②  $4x + 5y - 6 = 0$

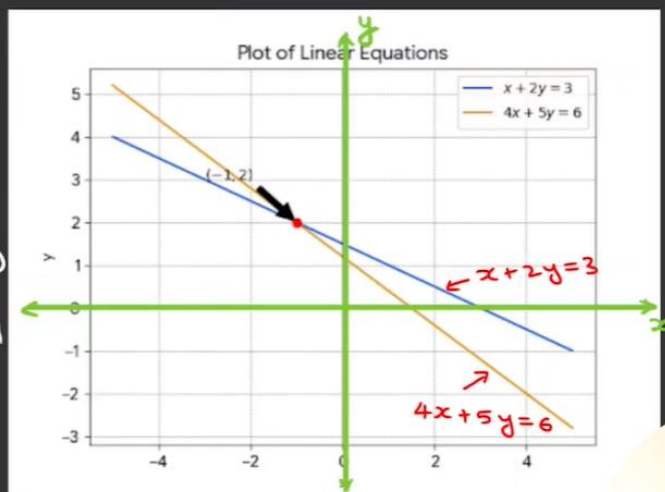
→  $x + 2y = 3$

$4x + 5y = 6$

↳ each equation represents infinite solution pair  $(x, y)$

↳ Equation ① and ② combined called system of linear equations

↳ sometimes we need  $(x, y)$  that satisfies each equation in systems of linear equations.



↳ Considering ① and ② together as a system of linear equations solution is

$(x, y) = (-1, 2)$

Example: Customer A purchased 2 apple and 2 banana for Rs 30 from a shop. Customer B purchased 3 apple and 1 banana for Rs 35 from same shop. what is price of apple and banana?

→ let apple price =  $x$   
banana price =  $y$

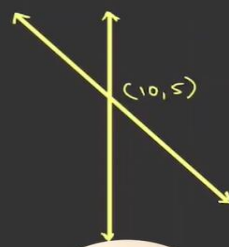
$2x + 2y = 30$  — ①

$3x + y = 35$  — ②

on solving ① and ②

$x = 10$   
 $y = 5$

Unique solution



Example: Customer A purchased 2 apple and 2 banana for Rs 30 from a shop. Customer B purchased 3 apple and 3 banana for Rs 45 from same shop. what is price of apple and banana?

→ let apple price =  $x$   
banana price =  $y$

$2x + 2y = 30$  — ①

$3x + 3y = 45$  — ②

on solving ① and ②

$x + y = 15 \Rightarrow (x, y) = (a, 15 - a)$ , where  $a$  is any real number.

Infinite solution possible

### ③ Linear equation in 3 Variables

$$ax + by + cz + d = 0 \leftarrow \text{Represents a plane in } 3D(R^3)$$

Example: ①  $2x + y + z = 5$

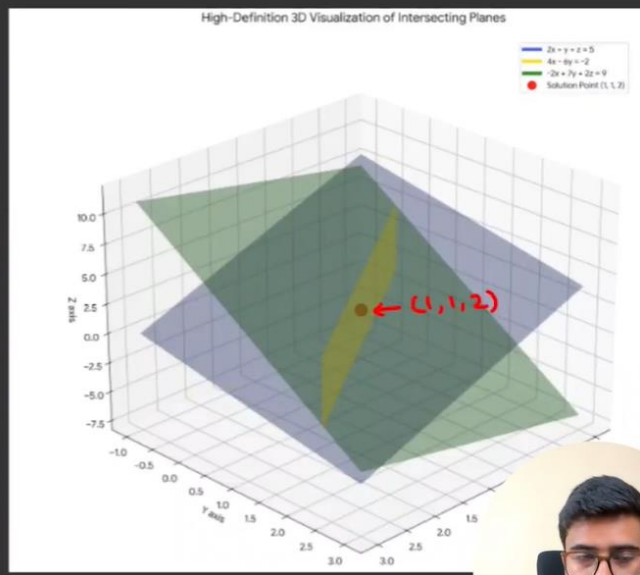
②  $4x - 6y = -2$

③  $-2x + 7y + 2z = 9$

→ Each equation represents a plane.

→ Geometrically, eq<sup>n</sup> ① and ② intersection represents a line

→ Intersection of line and plane ③ gives a point →  $(1, 1, 2)$



### ④ Linear equation in n variable

$$a_1x_1 + a_2x_2 + a_3x_3 + \dots + a_nx_n + b = 0$$

↑ Represents a  $(n-1)$  dimensional hyperplane in  $R^n$

#### \* Row Picture and column Picture

→ For a system of linear equation with n variable

↳ **Row Picture** consists of  $(n-1)$  dimensional hyperplane in n dimensional space.

↳ And if everything works out all plane meets at one point and that is the solution.

↳ Row picture is difficult to visualize and interpret geometrically

→ Let's look into Column picture.

Example:  $\underline{x + y = 5}$   
 $3x + y = 15$

Discuss the column picture

$$\rightarrow \begin{bmatrix} 1 & 1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 15 \end{bmatrix} \quad \leftarrow \text{Matrix form } Ax = b$$

↑  
Coefficient Matrix

↑  
vector of Unknown

↑  
Constant vector

$$\hookrightarrow x \begin{bmatrix} 1 \\ 3 \end{bmatrix} + y \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 15 \end{bmatrix} \quad \Rightarrow \quad \begin{matrix} x = 5 \\ y = 0 \end{matrix}$$

↑ Linear combination of 3 vectors and each one is a 3d vector. So we want to know for what value of  $x, y$  and  $z$ , their linear combination will give  $\begin{bmatrix} 5 \\ -2 \\ 9 \end{bmatrix}$

### \* Introduction to system of Linear equations

→ A system of Linear equations is simply a collection of Linear equations considered together, all involving the same variables.

**\*Typo**

→ A system of linear equations look like:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_n \end{cases}$$

**\*x1** **\*x2**

→ Each equation is a hyperplane in  $\mathbb{R}^n$

→ Solution of system of Linear equations means set of points that satisfy all equations simultaneously.

→ Let's look into column picture for above system of linear equations

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_n \end{bmatrix}$$

$$x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + x_3 \begin{bmatrix} a_{13} \\ a_{23} \\ \vdots \\ a_{m3} \end{bmatrix} + \dots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

### \* Machine Learning Application

→ We can represent Machine Learning problems as  $Ax = b$

→ Linear Regression Example:

| Area | #bedrooms | #bathroom | #floors | is_vip | Price |
|------|-----------|-----------|---------|--------|-------|
| —    | —         | —         | —       | —      | —     |
| —    | —         | —         | —       | —      | —     |
| —    | —         | —         | —       | —      | —     |

$$\hat{y} = w_1 \times \text{Area} + w_2 \times \text{bedrooms} + w_3 \times \text{bathrooms} + w_4 \times \text{floors} + w_5 \times \text{is\_vip}$$

↑ Linear Regression Model

↑ dummy dataset for housing Price Prediction (10000 Samples)



→ Linear Regression problems can be represented as

$$A x = b \leftarrow \text{labels}$$

data Matrix (features)  $\uparrow$   $\uparrow$  Parameters (weights + bias)

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{15} \\ a_{21} & a_{22} & a_{23} & \dots & a_{25} \\ \vdots & & & & \\ a_{10K,1} & a_{10K,2} & a_{10K,3} & \dots & a_{10K,5} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_{10K} \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_{10K} \end{bmatrix}$$